

STRUCTURE AND MECHANISM OF THE PLANT PIN-FORMED and AUX/LAX AUXIN TRANSPORTERS

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ABSTRACT

Auxin is a class of hormones that influences almost every aspect of plant growth and development. The AUX/LAX protein family mediates auxin import from the apoplast to the cytosol, whereas the PIN protein family mediates auxin export out of cells; both processes are essential for proper auxin distribution. The local auxin concentrations generated by both transporters function as organizing signals that govern tissue patterning and the establishment of organ polarity in plants. Auxin transporters shape polar auxin gradients through their substrate specificity, transport affinity, and regulated activity, including inhibition by endogenous and exogenous compounds (e.g., herbicides). For decades, the lack of structural and biochemical data has impeded a detailed comprehension of the molecular mechanisms behind auxin transport. In my talk, I will present recent biochemical, cryo-EM, and MD simulation data on the auxin importer AUX/LAXs and the auxin exporter PINs, which are essential for proper plant growth and tropisms. Our work provides a comprehensive and detailed transport mechanism, including a clear model for how auxin, herbicides, and competitive substrates are selected and what powers the transport.

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